

VU DINH HOANG

Hanoi, Vietnam

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SUMMARY

Backend-focused Software Engineering student with experience building distributed systems, scalable REST APIs, and AI-powered backend applications. Interested in system design, microservices architecture, and data-intensive backend systems involving databases, caching, and real-time processing.

SKILLS

Languages	Java, JavaScript, Python, SQL
Backend	Spring Boot, Node.js, REST APIs, WebSocket, gRPC
Databases	PostgreSQL, Redis, MongoDB, Neo4j
System & DevOps	Docker, Docker Compose, Nginx, GitHub Actions
Concepts	Distributed Systems, System Design, Microservices, Caching, Message Queues
Tools	Git, Postman, Swagger

PROJECTS

Voice Assistant AI System GitHub Feb 2026 – May 2026

Backend / AI Systems Developer

- Designed and built a distributed microservices system for voice-based AI interaction including STT, TTS, and LLM conversation
- Developed backend services using Node.js and Python (Flask) with PostgreSQL and Redis for data storage and caching
- Implemented real-time communication using WebSocket (Socket.IO) and asynchronous processing using BullMQ
- Designed LLM routing layer to optimize inference cost, latency, and model selection strategy
- Containerized system using Docker and deployed using CI/CD pipeline (GitHub Actions + Nginx reverse proxy)

TechRadar VN – Knowledge Graph & AI Backend System GitHub Sep 2025 – Nov 2025

Backend Developer

- Built backend services for an IT trend analysis platform using Spring Boot, PostgreSQL, Redis, and Neo4j
- Designed RESTful APIs for search, chatbot, analytics, and graph exploration features
- Developed a Graph RAG pipeline combining vector retrieval and knowledge graph traversal for question answering
- Built data collection and processing pipeline for Vietnamese technology sources
- Deployed containerized services using Docker Compose

Web Book – E-commerce Book Store GitHub January 2025 – March 2025

Full-stack Developer

- Built full-stack e-commerce system (Spring Boot + React) with JWT/OAuth2 authentication, MySQL, Redis, and payment integration
- Deployed scalable containerized architecture using Docker, Nginx, VNPay, and SendGrid

EDUCATION

VNU University of Science (HUS-VNU)
Bachelor of Computer Science

2023 – Present